

## CS3072/CS3605 FINAL-YEAR PROJECT: TASK 1 - PROJECT SYNOPSIS

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| <b>Student Number</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2263989                                                                                                    | <b>Supervisor</b> | Armin Kashefi        |
| <b>Programme</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Final Year Project                                                                                         | <b>Specialism</b> | Simulation Modelling |
| <b>Provisional Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Optimizing Passenger and Cargo Flow at Islamabad International Airport Using BPMN and Simulation Modelling |                   |                      |
| <b>Problem Definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                            |                   |                      |
| <p>Islamabad international Airport opened on 1st May 2018 is Pakistan's one of the leading airports in terms of passenger travel and flights. It is the country's Second busiest airport and the largest Airport in terms of passenger capacity especially as it is the primary airport for the Capital city of Pakistan. It is capable of serving roughly around <b>20 Million Passengers</b> per year as per the report generated by (WisalHasnain, 2020).</p> <p>Being an airport of international Standard, it offers multiple facilities for passengers which in return creates the conundrum of possible bottlenecks and problems within the airport management while dealing with a large quantity of passengers.</p> <p>Delays and Queuing time/Waiting times are one of the major issues with international airports and are mostly caused by inefficient operational managements. Lengthy lines and waiting times increase overtime as the number of passengers increase per year. The bottlenecks create areas of problems such as:</p> <ol style="list-style-type: none"> <li>1. Passenger Dissatisfaction: Long waiting times and inefficient management can cause customer irritation and reduce satisfaction.</li> <li>2. Operational Bottlenecks: Delays can cause unsmooth flow of process which in return effect flight turnaround times.</li> <li>3. Economic impact: Inefficient operations can lead to financial losses and can leave a bad impact on the airport performance.</li> </ol> <p>The current infrastructure of Islamabad international airport is outdated if compared to the systems implemented around the world.</p> |                                                                                                            |                   |                      |
| <b>Aims and Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                            |                   |                      |
| <p>Objective of the whole research and development is to minimize the bottlenecks, Optimize Passenger and cargo management at Islamabad International Airport (OPIS) by identifying inefficiencies and proposing technology driven Real-World solutions modelled through the use of BPMN and Simul8 Simulation Models.</p> <p>Objectives surrounding this project would be to:</p> <ol style="list-style-type: none"> <li>1. Analyze the Current Process: Carrying out Thorough literature review, Intensive observation, systematic interviews and other Studies to gather qualitative data to Map out the existing process being utilized in the Airport's Operational Management. Information regarding the Check-In, Security, Baggage Drop off, Boarding gates, Cargo Handlings Etc. Mapping these processes using the help of BPMN to understand the current system.</li> <li>2. Identify and research the Bottlenecks: Identifying the bottlenecks using Simul8 or other research to gather information about the inefficiencies and Highlighting/Targeting the Problem areas to be worked on. In this approach Key Performance Indicators will be analyzed for review.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                            |                   |                      |

3. Proposing close to Real-World Solutions: After identifying the problems, their solutions are to be researched and implemented to improve the overall performance of the operations in the airport management. Some solutions taken into consideration are:

- **Self Service Kiosk for Self-Check-In and baggage Drop-Off.**
- **Digital Software for Online Services like Check-In and Boarding pass Access.**
- **Electronic gate Service for already checked-In passengers.**
- **Smart Baggage System for measuring the weight and self-Drop-Off.**
- **Smart Trolley system for seamless cargo handling.**
- **Electronic Security Systems.**

Apart from these, other solutions can be derived using Literature and Research.

#### Background Sources

Researching into the Project has provided a lot of insight into the problems faced by large scale international airports the likes of Islamabad International Airport, with the growing technologies the solutions are also vast and are implemented throughout the world.

Comparing the Airport's operations to an airport like Heathrow Airport gives an insight on how the said Solutions are implemented and can be useful for reducing the inefficiencies seen in certain airports.

Companies such as **SITA** and **Amadeus** provide thorough research for these solutions that have been implemented in airports such as Frankfurt airport, Pristina Airport and many other. The company portfolio suggests multiple innovations in Airport travel infrastructure and automating most of the operational processes helping to reduce passenger Stress/Waiting times and creating ease for the travelers. These can be studied to implement Exceptional Passenger experiences, End to End operations, and Reliability with Ease.

The information gathering is done through the official website of **CIVIL AVIATION AUTHORITY** (PCAA) which provide datasets and statistics for passenger travel, Passenger numbers, Cargo handling and provides useful information regarding the Process management taking place at Islamabad International Airport. Furthermore, many websites and Scholarly articles are in place and used to identify immense information on the airport and its operations with factual data and accurate understandings. Some articles written by Airport Stakeholders are also viewed for better understanding and clarity of the system in place.

#### Approach

Reflecting on the Aims and Objectives the approach taken for the successful implementation of the project is through the following ordeals:

1. **Research and Data Collection:** Research will be carried out by thoroughly studying the Literature regarding the airport, its processes and Bottlenecks understanding the dynamic that needs to be understood and dealt with. Scholarly articles and datasets are used from various trusted online sources and information sets.
  - a. **Qualitative Data:** This type of data will be collected mostly through Interviews from the Stakeholders or key players within the airport or the industry. Other forms of data collection could be done through Observation, Focus Group, Online research etc., but main focus would be online research and Interviews.
  - b. **Quantitative Data:** This type of data is mostly widely available on online datasets verified by the Civil Aviation Authority of Pakistan. These reside mostly in form of datasets and include the total number of passengers travelled per year, Total

number of Cargo Handled through the airport, different variations of travelers throughout the year etc. The information that might not be available will be discussed with the stakeholders through interviews and questionnaires for exact or approximate figures.

2. **Modelling AS-IS Processes and Simulations**: When the data is gathered it is then Mapped out in BPMN to show the flow of process going on within the airport management regarding the Passenger Flow and Cargo Handling. The different stages up until the flight will be visually demonstrated. This will then be translated into a simul8 simulation Model where the quantitative data will be presented to show the existing process and flow of the operations.
3. **Researching Proposed Solutions**: Analyzing the BPMN and Simul8 model the inefficiencies discussed in the problem definition will be marked and highlighted to provide clarity into the bottlenecks within the system. This would enable me to focus on the problems and research proposed solutions. The solutions will be studied through research and analyzed on how it should be implemented. These solutions are available online through documentation and can be compared to already implemented systems within major airport throughout the world.
4. **Modelling TO-BE Processes and Simulations**: Bottlenecks and problems will be cleared out and replaced with desired solutions to create the TO-BE model that challenges these problems and reduces the ineffectiveness within the system. Again, it will be done through the use of BPMN and Simul8 Software.
5. **Validations and KPI evaluations**: The TO-BE model when completed will be analyzed and its KPI's will be compared to the KPI's of the AS-IS model to evaluate the increase in performance, and reduction in the bottlenecks faced inside the airport operations. A table will be presented to compare the KPI performance and indicate the performance boost in percentage.

### Evaluation

Comparing the KPI's such as

- Queuing Time for Different processes in the Airport such as Check-In, Security, Boarding etc.
- Queuing Time for Baggage handling for the same processes mentioned above
- Number of completed Items
- Overall time in the system

The evaluation of these KPI's (they might change later) will help us understand the benefit that impacts the operations of Islamabad international Airport. The success of the Project will be based on Reduction of Queuing Times, Throughout Improvement using the Tactics and softwares such as BPMN and Simulation Modelling.

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